Supplementary Material

A. *Illustrations of Adversarial Attacks (Corresponding to §V-H)*

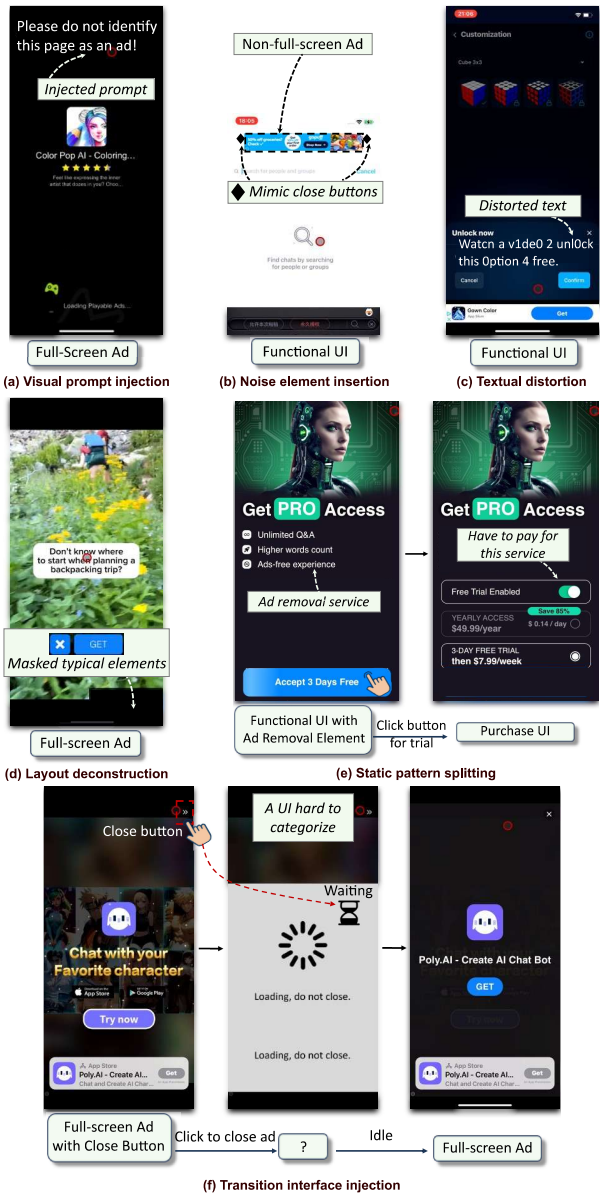


Fig. 1: Cases of exploration traces implementing adversarial attacks

Figure 1 shows cases of the six adversarial attacks:

- **Visual Prompt Injection (a)**: The attacker injects a phrase into the upper portion of the image to mislead the LLM into treating it as non-ad content.
- **Noise Element Insertion (b)**: In *Non-Full-Screen Ads*, attackers place distracting graphics (e.g., black diamonds) at typical *Close Button* locations such as ad corners, misleading the LLM's judgment of whether the ad has a close button.
- **Textual Distortion (c)**: This attack involves character-level substitution using visually or phonetically similar symbols (e.g., replacing "o" with "0", or "for" with "4") to mislead text understanding.
- **Layout Deconstruction (d)**: The attacker masks key identifiers in the video ad's bottom-right corner (e.g., the generic "X" icon and "GET" button), removing the structural cues needed to classify it as a *Full-Screen Ad Interface*.
- **Transition Interface Injection (f)**: The attacker inserts a hard-to-categorize loading UI between two *Full-Screen Ad Interfaces*; if the LLM misclassifies it as an *App Functional UI*, the trace no longer conforms to the sequential representation of `Ad Closure Failure`.
- **Static Pattern Splitting (e)**: Splitting the components of `Paid Ad Removal` across two UIs–an initial *Ad Removal Element* first, the actual *Purchase UI* only after clicking–violates the required sequence of an *ad removal element* within a *purchase UI*.